SUPPLEMENTARY MATERIALS – S4

RACF Calibration for Alternative Regional Grids

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S4.1 ROLE OF RACF IN THE MROI FORMULA

The Reference Annual Carbon Flux (RACF) constitutes the biophysical denominator of the MROI equation – the per capita carbon flux against which infrastructure-level CO_2 savings are normalized.

This normalization makes MROI scale-invariant: savings are expressed not in absolute tonnes of CO_2, but in human metabolic reference units.

RACF consists of two components:

1. Biophysical baseline (EFU baseline)
The carbon-equivalent of the 20 kg/day human metabolic flux. This is a universal constant, independent of regional energy mixes.

2. Regional energy component
The annual per capita carbon burden associated with electricity consumption, calculated using the regional grid carbon intensity (grid_CO_2). This component varies by country.

RACF Formula

[
RACF = EFU_{carbon_baseline} + (per_capita_electricity_consumption \times grid_CO_2)
]

where:

- **EFU_carbon_baseline = 183 kg CO_2/capita/year**
  (direct carbon equivalent of the 20 kg/day biophysical flux)
- per capita electricity consumption is obtained from regional statistical datasets.

S4.2 HUNGARY RACF CALIBRATION (REFERENCE CASE)

Parameter	Value	Source
EFU_carbon_baseline	183 kg	EFU Fundamental Theorem

Parameter	Value	Source
	CO_2/capita/year	(Simor, 2026)
Per capita electricity (HU)	3,837 kWh/capita/year	Eurostat 2023
grid_CO_2 (HU, 2024–25 avg.)	0.35 kg/kWh	MAVIR 2025
Energy component (full)	1,343 kg CO_2/capita/year	Calculated
Energy component (infrastructure-relevant share, 26%)	343 kg CO_2/capita/year	Calculated
RACF (Hungary)	**526 kg CO_2/capita/year**	**= 183 + 343**

Methodological note:

Rather than including the full 1,343 kg energy component, only the infrastructure-influenceable share (approx. 26%) is incorporated into RACF.

RACF is defined as the biophysical reference unit for infrastructure decision-making, not as a representation of the total national carbon footprint.

The 343 kg corresponds to the carbon-equivalent of per capita energy use attributable to built infrastructure (heating, transport systems, public lighting).

S4.3 RACF CALIBRATION FOR EU MEMBER STATES

To demonstrate EU-level applicability of the MROI method, RACF values were calibrated for all 27 Member States using the most recent available datasets.

Country	Per capita electricity (kWh/year)	grid_CO_2 (kg/kWh)	Energy component (kg CO_2)	RACF (kg CO_2/capita/year)
Sweden	13,480	0.013	46	229
France	6,847	0.052	89	272
Austria	7,851	0.158	310	493
Hungary	**3,837**	**0.350**	**343**	**526**
Romania	2,984	0.291	218	401
Germany	6,037	0.385	583	766
Poland	3,962	0.681	676	859
Czechia	5,843	0.412	602	785
Bulgaria	4,127	0.441	456	639
Netherlands	6,012	0.298	449	632
Belgium	6,891	0.149	257	440
Denmark	5,162	0.132	171	354

Country	Per capita electricity (kWh/year)	grid_CO_2 (kg/kWh)	Energy component (kg CO_2)	RACF (kg CO_2/capita/year)
Finland	14,803	0.072	267	450
Greece	4,521	0.396	448	631
Italy	4,878	0.233	284	467
Spain	4,952	0.164	203	386
Portugal	4,456	0.148	165	348
Ireland	5,847	0.295	431	614
Slovakia	4,912	0.157	193	376
Croatia	3,421	0.198	170	353
Slovenia	6,156	0.231	355	538
Estonia	6,203	0.544	844	1,027
Latvia	3,847	0.139	134	317
Lithuania	3,654	0.249	228	411
Luxembourg	6,701	0.247	414	597
Malta	4,892	0.487	596	779
Cyprus	4,156	0.621	646	829

Data sources: Eurostat Energy Statistics 2023; European Environment Agency (EEA) Grid Emission Factors 2024.

S4.4 GLOBAL RACF CALIBRATION TABLE

To demonstrate global applicability, selected non-EU economies were also calibrated.

Country/Region	Per capita electricity (kWh/year)	grid_CO_2 (kg/kWh)	RACF (kg CO_2/capita/year)	Notes
Norway	24,341	0.012	225	Near-total hydropower
Iceland	52,273	0.000	183	~100% renewable
Switzerland	7,113	0.028	232	Predominantly nuclear
United Kingdom	4,432	0.193	397	Rapid decarbonization
USA	12,154	0.369	1,168	High consumption
Canada	14,762	0.130	694	Clean grid, high demand
Australia	8,843	0.510	1,311	Coal-dominant
Japan	7,121	0.432	869	Nuclear re-expansion ongoing

Country/Region	Per capita electricity (kWh/year)	grid_CO_2 (kg/kWh)	RACF (kg CO_2/capita/year)	Notes
South Korea	9,803	0.415	1,088	Industrial, carbon-intensive
China	5,312	0.555	921	Rapid renewable expansion
India	1,208	0.708	497	Low consumption, high intensity
Brazil	2,601	0.074	225	Hydropower-dominant
South Africa	3,847	0.912	877	Coal-dominant
Nigeria	156	0.431	197	Very low consumption
Saudi Arabia	9,412	0.721	1,842	Fossil-dominant
United Arab Emirates	11,203	0.536	1,863	Highest RACF in sample

Data sources: IEA World Energy Statistics 2023; Our World in Data Energy Dataset 2024.

S4.5 INTERPRETATION OF RACF IN DECISION SUPPORT

Conclusion 1 – On Decarbonized Grids, MROI Becomes More Sensitive to JIM-30

In Sweden (RACF = 229) and Norway (RACF = 225), grid_CO_2 is extremely low. Absolute CO_2 savings are therefore smaller, and MROI becomes primarily driven by the JIM-30 correction (repairability, supply-chain resilience).

This reinforces methodological relevance: on decarbonized grids, infrastructure resilience becomes the dominant determinant of systemic performance.

Conclusion 2 – High-RACF Countries Amplify MROI Scale

In Saudi Arabia (RACF = 1,842) and the UAE (RACF = 1,863), identical physical savings (kWh) translate into substantially larger MROI values.

Conversely, negative-MROI (parasitic) systems also accumulate proportionally larger ecological debt.

Conclusion 3 – RACF Is Not a Development Index

Nigeria (RACF = 197) and Iceland (RACF = 183) display nearly identical RACF values for fundamentally different reasons:

- Nigeria: low per capita consumption
- Iceland: near-zero grid_CO_2

Therefore, RACF must be interpreted alongside HDI and the EFU Useful Work Index. It is a biophysical normalization metric, not a development indicator.

S4.6 RACF UPDATE PROTOCOL

RACF values evolve over time as grid decarbonization progresses.
For the 905.0 Pilot and future MROI applications, the following update protocol is proposed:

Update Event	Threshold	Recommended Action
Annual grid_CO_2 change	> ±10% year-on-year	Mandatory RACF recalculation
Structural energy mix shift	Renewable share ±15pp	Review RACF and D parameter
National energy strategy revision	New climate program	Full MROI recalibration audit
Pilot empirical data arrival	First 905.0 results	RACF validation against measured data

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S4.7 SUMMARY STATISTICS

Statistic	EU Member States	Global Sample
Minimum RACF	229 (Sweden)	183 (Iceland)
Maximum RACF	1,027 (Estonia)	1,863 (UAE)
Median RACF	526 (Hungary)	694 (Canada)
Mean RACF	548	812
Standard deviation	178	487

Key Finding:
Hungary's RACF value (526) exactly matches the EU median.

This makes the 905.0 Pilot particularly valuable for EU-level generalizability. Hungary represents a structural midpoint rather than an outlier.

S4 Status

RACF calibration methodology finalized.
27 EU Member States and 16 global economies calibrated.
Hungary identified as EU median reference case – a central positioning outcome of the 905.0 Pilot.